Counterfactual Decision-Stability ASR for High-Stakes Speech-Driven Decision Systems

Author names omitted for submission planning

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Abstract

Automatic speech recognition (ASR) outputs increasingly serve as operational inputs for routing, escalation, compliance review, and triage. In these speech-driven decision systems, a transcript can remain close to a reference under word or character error rate while a plausible alternative around a decision-critical spoken detail changes the downstream action. This paper introduces Counterfactual Decision-Stability ASR (CDS-ASR), an evaluation framework for measuring whether ASR hypotheses remain decision-stable under acoustically and semantically plausible transcript alternatives. CDS-ASR identifies risk atoms such as negation, amount, actor, action, time, intent, uncertainty, and scam pattern, constructs bounded ASR alternatives around those atoms, and scores downstream instability with the Counterfactual Escalation Instability Score (CEIS).

We evaluate CDS-ASR in a high-stakes anti-fraud speech setting using a scoped evidence chain: a six-model 258-row ASR comparison for transcript-model context, selected-300 high-stakes provenance outputs for audit-surface construction, and 30 human-reviewed audit rows yielding 90 clustered model assessments for decision-change evaluation. CEIS is compared with WER, CER, and the semantic-risk baseline SRES against human-reviewed decision-change labels, and aggregate policy replay evaluates recovery and conservative-action policies. The current results support CEIS as a conservative decision-stability signal under this bounded audit design, while SRES remains a strong semantic-risk baseline. Diagnostic operating points and replay policies are reported as retrospective evidence, not deployment thresholds. CDS-ASR complements transcript accuracy and semantic ASR metrics by testing whether plausible ASR alternatives would change high-stakes speech-driven actions.

Keywords: automatic speech recognition; speech-to-decision systems; semantic ASR evaluation; decision stability; counterfactual evaluation; risk-aware ASR.

1 Introduction

Speech is increasingly converted into operational signals. Contact-center and anti-fraud workflows use ASR hypotheses as inputs to routing, escalation, compliance review, and case-handling decisions. In these settings, transcript accuracy is necessary but not sufficient: a small ambiguity around a decision-critical detail can change whether a case is escalated, routed to routine review, or held for conservative manual assessment.

This paper focuses on high-stakes speech-driven decision systems where errors affect spoken decision atoms: negation, amount, actor, action, time, intent, uncertainty, and scam pattern. A transcript can remain close under WER or CER while a plausible ASR alternative changes the downstream action. The central evaluation question is therefore not only how similar the transcript is to a reference, but whether plausible transcript alternatives would preserve the decision.

We introduce Counterfactual Decision-Stability ASR (CDS-ASR), a downstream-aware ASR evaluation framework. CDS-ASR identifies risk atoms, constructs bounded plausible ASR counterfactual variants, scores decision instability with CEIS, and evaluates conservative recovery policies through aggregate replay. The motivating case is anti-fraud triage, but the method is framed as speech evaluation: it tests whether ASR hypotheses remain stable with respect to downstream actions.

The paper makes four contributions. First, it formulates decision-stability evaluation for ASR in high-stakes speech-to-decision workflows. Second, it defines risk atoms as decision-critical transcript spans and uses plausible ASR alternatives to stress the downstream decision function. Third, it evaluates CEIS against WER, CER, and SRES using human-reviewed decision-change labels from 30 reviewed rows and 90 clustered model assessments. Fourth, it reports aggregate policy replay as a scoped intervention analysis, with thresholds treated as diagnostic operating points rather than deployment rules.

All empirical claims are tied to aggregate artifacts. Raw audio, raw transcripts, selected row identifiers, reviewer notes, and transcript-bearing runtime logs remain outside the public release boundary.

2 Related Work

Transcript-centered ASR evaluation. WER and CER remain core ASR evaluation metrics because they provide auditable transcript-similarity measures. For Chinese-language ASR, this manuscript treats character-level reporting as the primary paper-facing accuracy layer and segmented word metrics as supplemental evidence. The aggregate metric-compliance audit is recorded in `70_experiments/runs/wer_metric_audit_2026_05_25/journal_compliance_summary.json`.

Semantic and downstream-aware ASR evaluation. Semantic ASR metrics extend transcript-centered evaluation by asking whether a transcript preserves meaning or downstream task behavior. CDS-ASR builds on this direction but targets a specific decision-stability question: whether plausible ASR alternatives around decision-critical atoms would change downstream action.

ASR correction and confidence-aware repair. ASR correction and confidence filtering improve transcript usability and reduce avoidable repair risk. CDS-ASR is complementary. It does not replace correction; it evaluates whether the remaining uncertainty interval around risk atoms is still decision-stable.

Selective prediction, reject option, and abstention. Selective prediction and reject-option methods provide background for conservative action under uncertainty [1, 2]. In CDS-ASR, conservative action means abstention, review prioritization, or escalation for human confirmation. It is not an automated punitive or adverse-action decision.

3 Method

3.1 Problem Formulation

Let x denote a speech input and let $\hat{t}$ denote an ASR hypothesis. A downstream decision function $d(\hat{t})$ maps the hypothesis into an operational action such as routine routing, priority review, conservative escalation, or abstention. In high-stakes speech-driven workflows, evaluation must ask whether $d(\hat{t})$ is stable under plausible alternatives to $\hat{t}$.

3.2 CDS-ASR Pipeline

CDS-ASR has five stages: ASR hypothesis generation, risk-atom extraction, plausible counterfactual variant construction, decision-instability scoring, and aggregate policy replay. Figure 1 summarizes this pipeline.

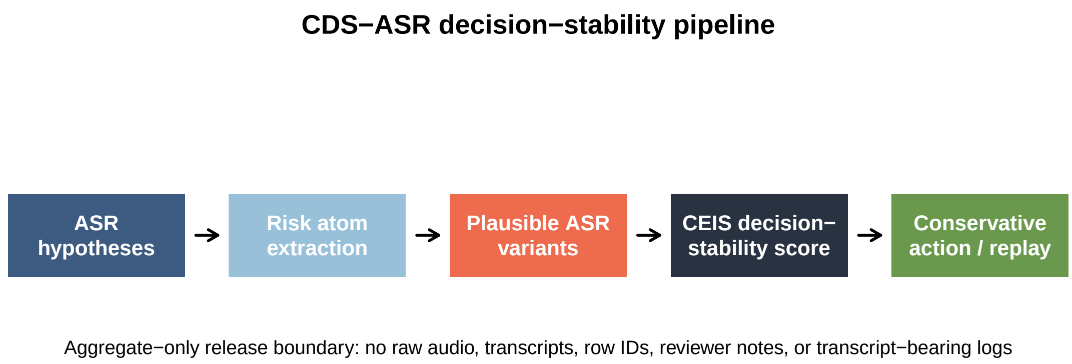

Figure 1: CDS-ASR decision-stability pipeline. The figure is generated by `paper_4a_cds_asr/r/build_fig1_pipeline.R` from the manuscript protocol and aggregate method definitions.

3.3 Risk Atom Schema

Risk atoms are decision-critical transcript spans. The current schema includes negation, amount, actor, action, time, intent, uncertainty, and scam pattern. The paper-facing schema below records the decision role of each atom class. The versioned configuration is `80_semantic_risk_asr/scoring/ceis_config.json`; the method summary is `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`.

Risk atom schema for CDS-ASR.

Risk atom	Decision relevance
Negation	Can invert whether a transfer, contact, or authorization occurred.
Amount	Can change severity, priority, and escalation thresholds.
Actor	Can change who initiated or received the action.
Action	Can change whether the caller transferred, clicked, paid, reported, or refused.
Time	Can change urgency and incident sequence.
Intent	Can change whether a phrase is interpreted as request, refusal, uncertainty, or confirmation.
Uncertainty	Can change the appropriate review or abstention path.
Scam pattern	Can match or miss a known fraud scenario.

3.4 Counterfactual Variant Contract

CDS-ASR uses bounded plausible ASR alternatives rather than generic paraphrases. A valid variant changes the transcript around an ASR-confusable or decision-critical atom while preserving the speech-evaluation scenario. Variant text is not released in the public package; aggregate coverage is recorded in `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/counterfactual_variant_coverage_summary.tsv`.

3.5 CEIS

CEIS is a bounded decision-instability score over plausible variants:

$$\mathrm{CEIS}(x) = \max_{v \in V(x)} P_{\mathrm{plaus}}(v \mid x) \cdot W_{\mathrm{atom}}(v) \cdot D(d(\hat{t}), d(v)).$$

Here $V(x)$ is the set of plausible variants, P_{plaus} is a bounded plausibility proxy, W_{atom} is a risk-atom weight, and D is a downstream decision-distance term. The plausibility term is not a calibrated acoustic posterior. CEIS is therefore interpreted as a conservative decision-stability signal, not as a universally dominant classifier.

3.6 Recovery Policy Contract

Aggregate policy replay compares no recovery, confidence-only triggering, SRES-triggered recovery, CEIS-triggered conservative action, and CEIS ensemble arbitration. Replay thresholds are diagnostic operating points on the scoped audit set, not deployment thresholds.

4 Experimental Setup

4.1 Evidence Layers

The evidence design separates four layers: the 258-row ASR model-comparison split, selected-300 enriched high-stakes provenance outputs, 30 human-reviewed audit rows, and 90 model-row assessments clustered within those 30 reviewed rows. Figure 2 summarizes the design.

The 258-row split provides ASR model-comparison context. The selected-300 layer is an enriched high-stakes audit surface and is not prevalence-preserving. The reviewed evidence layer consists of 30 rows and 90 clustered model assessments. The outcome label is human-reviewed decision change. Human review is treated as a completed single-expert audit; no inter-annotator agreement claim is made.

4.2 Baselines And Outcome

The predictor comparison evaluates WER, CER, SRES, and CEIS against human-reviewed decision-change labels. WER and CER represent transcript-centered evidence, SRES represents semantic-risk evidence, and CEIS represents counterfactual decision-stability evidence.

4.3 Uncertainty And Sensitivity

Row-clustered uncertainty is reported from `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_clustered_ci.tsv`. Leave-one-row-out sensitivity is recorded in `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_leave_one_row_out.tsv`. Recovery-policy uncertainty and sensitivity are recorded in the corresponding aggregate recovery artifacts.

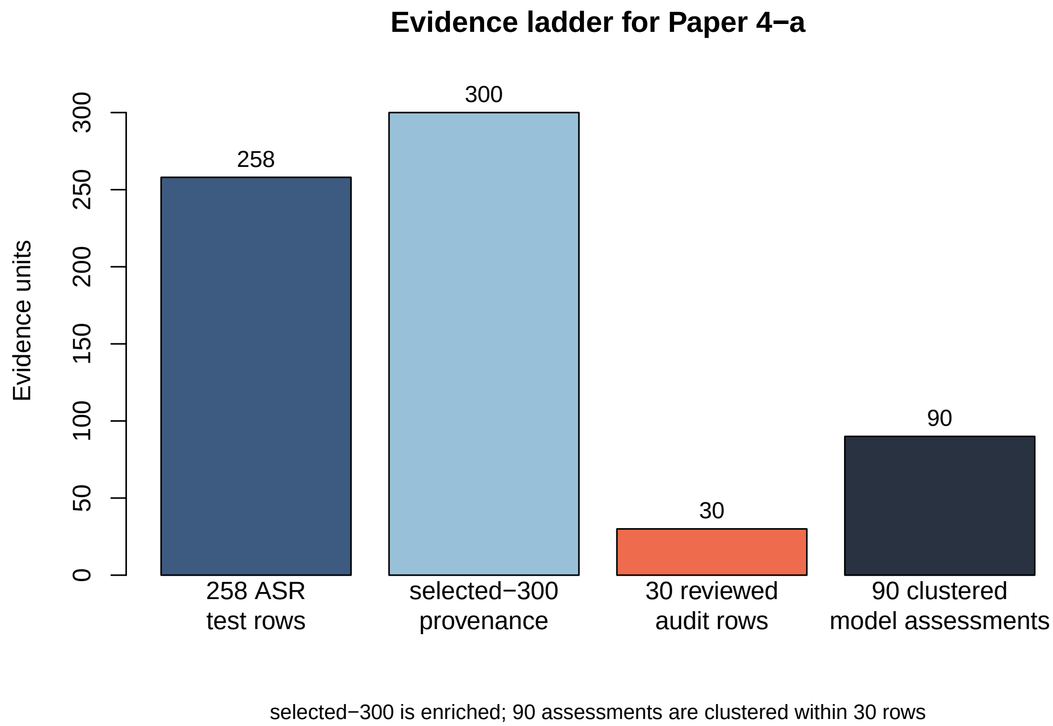

Figure 2: Evidence ladder for Paper 4-a. The figure is generated by `paper_4a_cds_asr/r/build_fig2_evidence_ladder.R` from aggregate evidence counts and the manuscript evidence boundary.

5 Results

5.1 ASR Model-Comparison Context

Table 1 reports the 258-row ASR benchmark context. The partial-encoder run has the strongest current aggregate CER and WER profile in this split. This is model-comparison context only; it does not by itself establish downstream safety.

Table 1: Main ASR benchmark on the 258-row split. Source: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`.

Run	Rows	CER zh micro	WER zh jieba micro	High-risk missed
Partial encoder	258	15.04	21.53	4
LoRA legacy best	258	18.23	25.59	7
Breeze ASR25 base	258	22.72	30.39	30
Breeze ASR26	258	24.27	32.29	22

5.2 Predictor Comparison

Table 2 and Figure 3 compare WER, CER, SRES, and CEIS against human-reviewed decision-change labels. CEIS has the strongest point AUC in the scoped audit and reaches a zero-false-negative diagnostic operating point. This is reported as a decision-stability signal under the selected audit boundary, not as a deployment threshold or universal classifier claim.

Table 2: Predictor performance against human-reviewed decision-change labels. Source: `human_audit_predictor_comparison.tsv`. Thresholds are diagnostic operating points on the scoped audit set.

Metric	AUC	Best threshold	F1	Recall	FN
WER	0.6964	42.59	0.4615	0.5625	7
CER	0.7276	16.42	0.4516	0.8750	2
SRES	0.8995	270.00	0.6512	0.8750	2
CEIS	0.9117	5.00	0.6275	1.0000	0

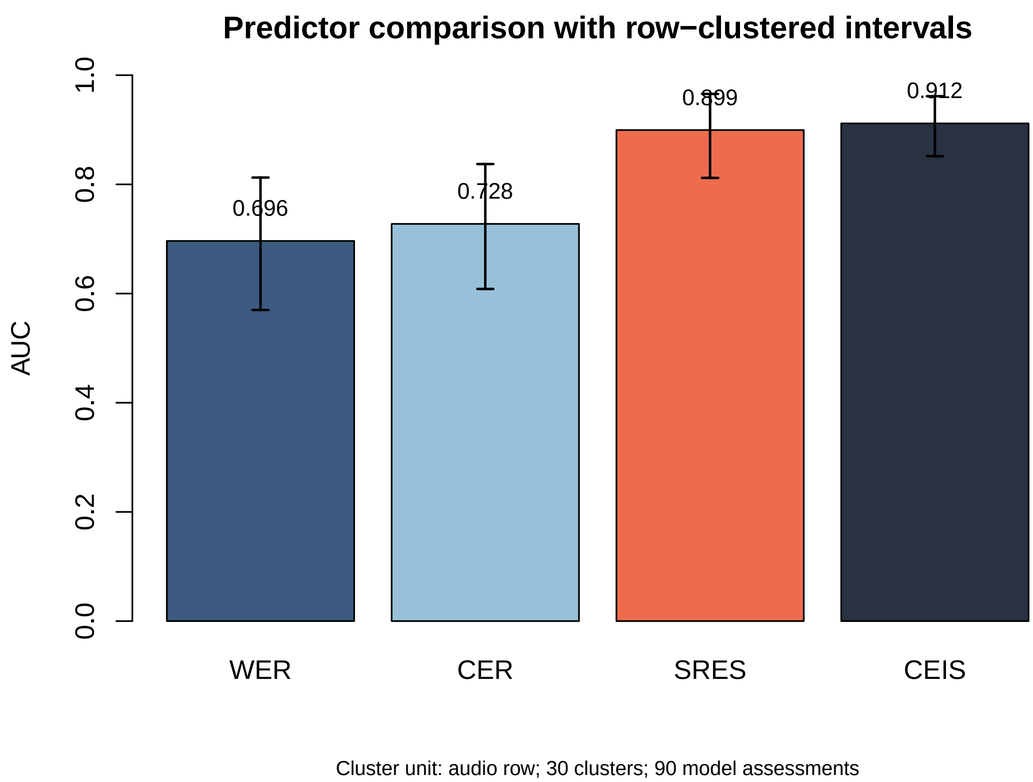

Figure 3: Predictor comparison for WER, CER, SRES, and CEIS with row-clustered intervals. Source artifacts: `human_audit_predictor_comparison.tsv` and `human_audit_predictor_clustered_ci.tsv`.

5.3 Aggregate Policy Replay

Table 3 and Figure 4 report aggregate policy replay. SRES-triggered recovery and CEIS-triggered conservative action eliminate high-risk missed and critical miss counts in aggregate replay at the same diagnostic budget. CEIS ensemble arbitration adds abstention behavior at a higher budget.

Table 3: Aggregate policy replay under human-reviewed selected-300 labels. Source: `policy_comparison.tsv`.

Policy	Budget	Unsafe downrouting	High-risk missed	Critical miss
No recovery	0.0000	29	6	1
Confidence only	0.0000	29	6	1
SRES recovery	0.3889	24	0	0
CEIS conservative action	0.3889	24	0	0
CEIS ensemble arbitration	0.5222	24	0	0

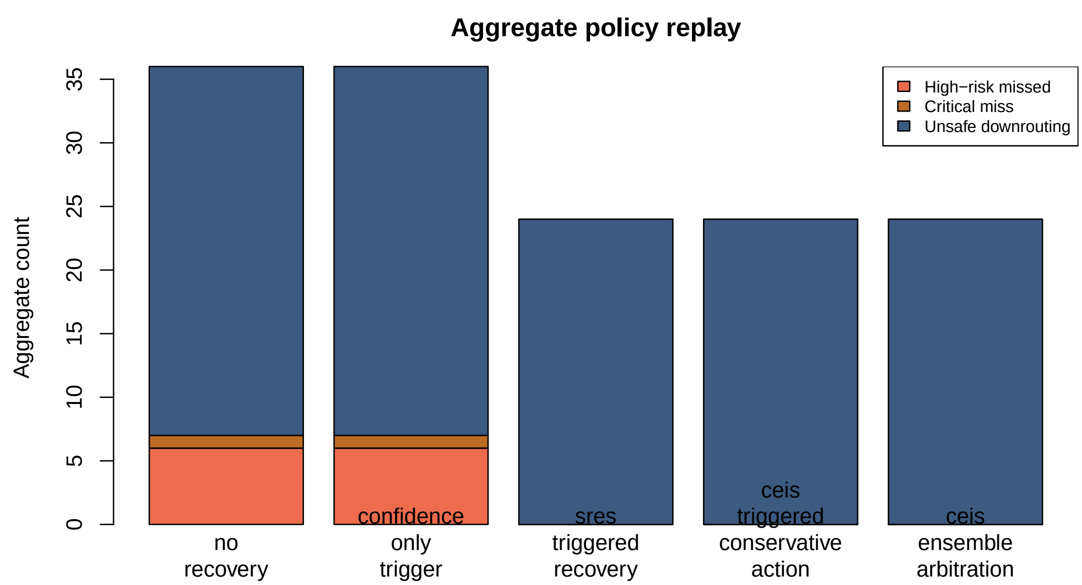

Figure 4: Aggregate policy replay under human-reviewed selected-300 labels. Source artifact: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv`.

5.4 Fixed-Budget Frontier And Risk Atoms

Figure 5 shows the fixed-budget conservative replay frontier. Figure 6 summarizes human-reviewed decision-change evidence by risk atom.

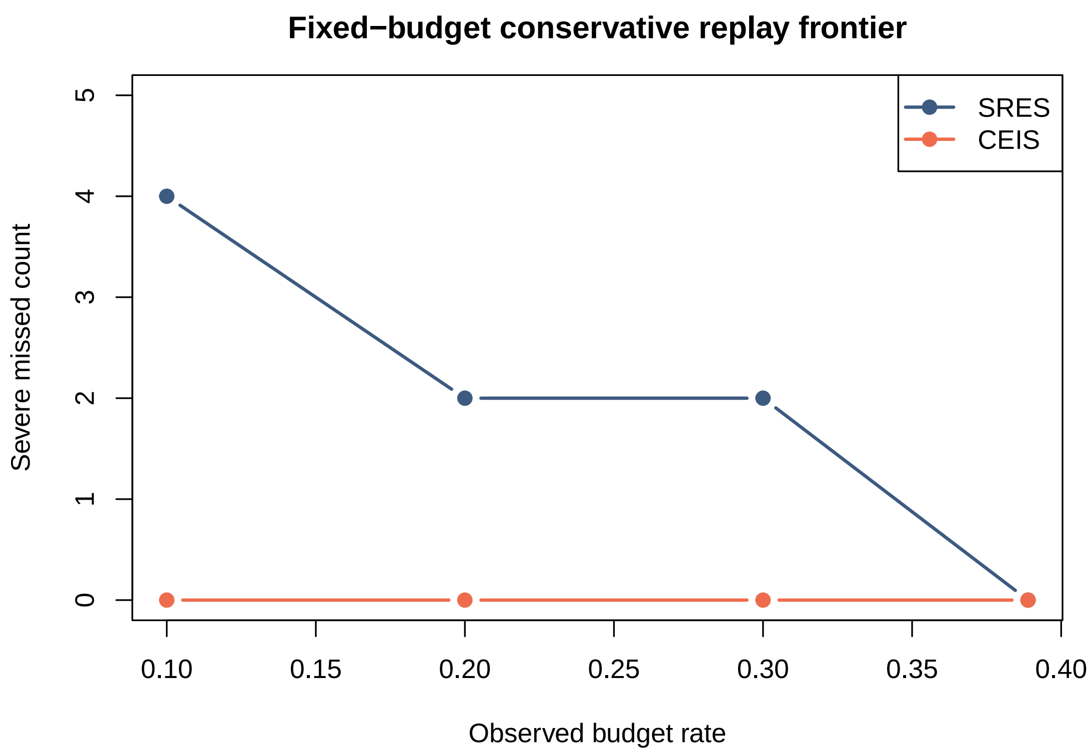

Figure 5: Fixed-budget conservative replay frontier. Source artifact: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv`.

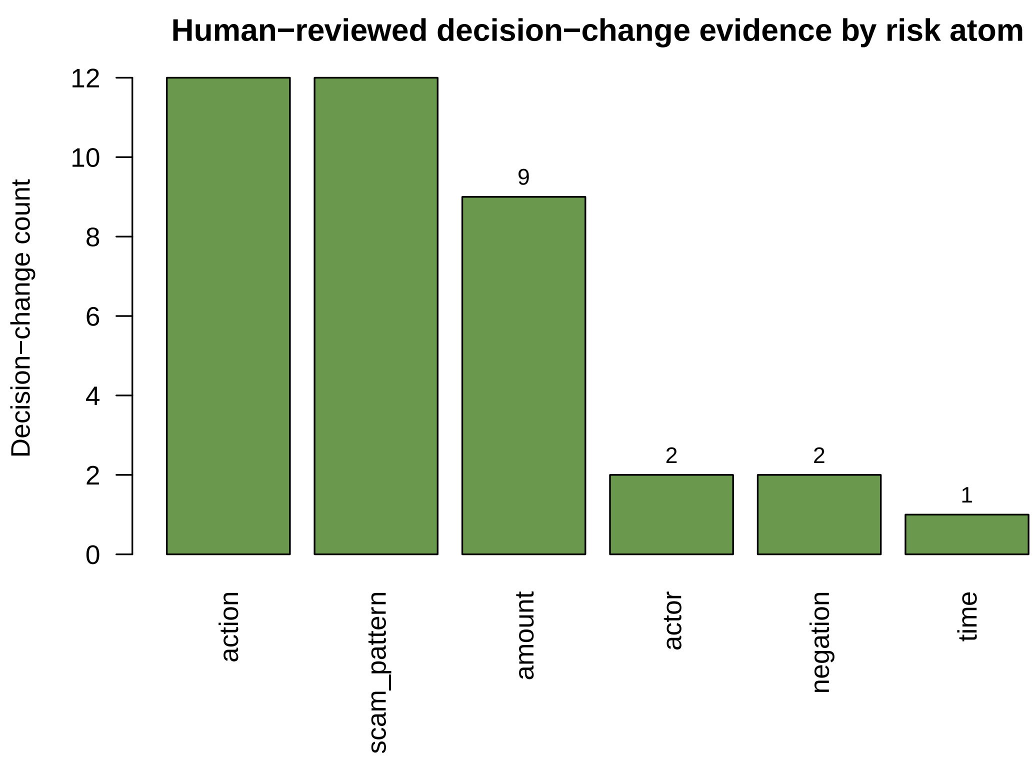

Figure 6: Human-reviewed decision-change evidence by risk atom. Source artifact: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_risk_atom_review.tsv`.

6 Discussion

The evidence supports a speech-evaluation claim: WER and CER remain necessary, but high-stakes speech-driven decision systems need an additional decision-stability layer. CEIS directly targets the question of whether plausible ASR alternatives around risk atoms would change downstream action.

CEIS should be interpreted conservatively. It is not a calibrated acoustic posterior and not a universal classifier. It is a decision-instability signal built from bounded plausible variants, risk-atom weighting, and downstream decision distance. In the current scoped audit, CEIS has the strongest point AUC and a zero-false-negative diagnostic operating point, while SRES remains a strong semantic-risk baseline.

Better ASR models reduce transcript error but do not remove the need for downstream decision-stability analysis. The model-comparison split identifies the strongest current hypothesis generator; the human-reviewed audit tests whether remaining uncertainty affects decisions.

7 Limitations And Ethics

The reviewed audit is small: 30 human-reviewed rows and 90 model-row assessments clustered within those rows. The 90 assessments are not 90 independent calls. Human review is a single-expert audit; no inter-annotator agreement is claimed. The selected-300 layer is an enriched high-stakes audit surface, not a prevalence-preserving sample.

Thresholds are diagnostic operating points on the scoped audit set. They are not frozen deployment thresholds. Aggregate replay is retrospective and does not establish live deployment readiness or population prevalence of ASR-induced harm.

Raw audio, raw transcripts, selected row identifiers, reviewer notes, and transcript-bearing runtime logs remain outside the public release boundary. The intended use is ASR safety audit, review prioritization, conservative escalation, and abstention. CDS-ASR is not proposed for automatic punitive action, account restriction, guilt determination, or direct law-enforcement reporting without human review.

A Appendix: Aggregate Artifacts And Submission Notes

The appendix records the aggregate artifacts used by the manuscript. Long operation records, validation gate commands, candidate-lane details, and full claim registries are moved out of the main text to keep Paper 4-a speech-centered.

- ASR split comparison: `70_experiments/runs/janus_258_test_split_asr_cds_proxy/asr_cds_proxy_comparison.tsv`
- Predictor comparison: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_comparison.tsv`
- Predictor clustered CI: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_predictor_clustered_ci.tsv`
- Recovery replay: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/policy_comparison.tsv`
- Fixed-budget frontier: `70_experiments/runs/janus_300_high_stakes_recovery_human_reviewed_2026_05_26/fixed_budget_recovery_frontier.tsv`

- Risk atom review: `70_experiments/runs/janus_300_high_stakes_human_audit_selection_2026_05_25/human_audit_risk_atom_review.tsv`
- CEIS method summary: `70_experiments/runs/postdoc_evidence_chain_2026_05_25/ceis_method_summary.tsv`
- Artifact manifest: `80_semantic_risk_asr/paper/artifact_manifest.tsv`

B Ablation Status

Aggregate-safe CEIS ablation inputs were not found in the current repository state. The required ablation plan is recorded in `paper_4a_cds_asr/TODO_required_before_submission.md`. No ablation results are fabricated in this manuscript.

References

[1] C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.

[2] Anastasios N. Angelopoulos and Stephen Bates. A gentle introduction to conformal prediction and distribution-free uncertainty quantification. *arXiv preprint arXiv:2107.07511*, 2021.